KENYA MEDICAL TRAINING COLLEGE
FACULTY OF CLINICAL SCIENCES
DEPARTMENT OF ORTHOPAEDIC AND TRAUMA MEDICINE

FINAL QUALIFYING EXAMINATION
FOR
CERTIFICATE IN ORTHOPAEDIC AND TRAUMA MEDICINE

PAPER: INTRODUCTION TO TRAUMATOLOGY

DATE: 3RD FEBRUARY 2025	TIME: 3 HOURS (2:00PM – 5:00PM)

INSTRUCTIONS

1. This paper consists of:
   - Section 1 (40 Multiple Choice Questions)
   - Section 2 (8 Short Answer Questions)
   - Section 3 (1 Long Answers Question)
2. Attempt **ALL** Questions
3. **ALL** Multiple choice questions **MUST** be answered in **CAPITAL LETTERS**
4. Write your **EXAMINATION NUMBER** on all the answer sheets provided and on the question paper.
5. No other writing should be done on the question paper except your Examination Number.
6. Ensure that all examination answer scripts are handed in at the end of the examination
7. Ensure you sign the examination register provided

EXAMINATION NUMBER .....

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SECTION 1: MULTIPLE CHOICE QUESTIONS (40 MARKS)

1. A 28year rugby player fell while playing and sustained injury to the right shoulder. on examination the upper arm is externally rotated, slightly abducted and loss of shoulder contour. The most likely diagnosis for this patient is?
   A. Posterior shoulder dislocation
   B. Central shoulder dislocation
   C. Anterior shoulder dislocation
   D. Recurrent shoulder dislocation
2. In Galeazzi fracture dislocation, the forearm should be immobilized in which position:
   A. Pronation
   B. Mid pronation
   C. Supination
   D. Mid supination
3. The commonest displacement in supracondylar humeral fracture is?
   A. Anterior
   B. Medial
   C. Dorsal
   D. Lateral
4. You are informed that a patient with Webber C injury has been admitted in the ward. This patient has injury of?
   A. Shoulder joint
   B. Knee joint
   C. Elbow joint
   D. Ankle joint
5. Which one of the following characterizes Gartland classification of supracondylar fractures?
   A. Gartland II is worst in all
   B. Gartland III can do without surgery
   C. Gartland I can heal on a cast
   D. Gartland I is the worst and needs surgery
6. which one of the following best describes injuries of the wrist?
   A. Lunate bone often fractures at its neck
   B. Dynacast works better in wrist treatment
   C. Scaphoid often affected by avascular necrosis
   D. Scaphoid injury the most rare
7. Which of the following fractures is associated with haematuria?
   A. Neck femur fracture
   B. Intertrochanteric fracture
   C. Pelvic fracture
   D. Lumber fractures

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8. Which of the following is confirmatory clinical feature of a fracture?
   A. Swollen limb
   B. Tenderness
   C. Shouting patient
   D. Crepitus
9. Which of the following is a feature of green stick fractures?
   A. Occurs mostly to senile patients
   B. Common of college students
   C. Fracture only in rickets children
   D. Occurs to children with resilient bones
10. Anterior shoulder dislocation can result from:
   A. Anterior force to the Victim
   B. Medial force with lateral rotation
   C. Vertical compression
   D. Posterior trauma force
11. Which nerve is injured following a fracture of the mid shaft humerus?
   A. Axillary nerve injury
   B. Ulna nerve injury
   C. Radial nerve injury
   D. Median nerve injury
12. In posterior dislocation of the shoulder, electric bulb appearance is seen in which X-ray view?
   A. Axillary view
   B. Postero-Anterior view
   C. Antero-posterior view
   D. Lateral view
13. Carpal tunnel syndrome is common in fractures of?
   A. Children femur fractures
   B. Fracture proximal humenis
   C. Distal fracture radius
   D. Fracture neck humenis
14. Which one of the following is an indication of external fixation?
   A. Multiple fractures
   B. Fracture in elderly patient
   C. Fracture neck femur
   D. Compound fractures
15. Which of the following is the treatment of Stress fracture?
   A. Internal fixation
   B. Rest
   C. Closed reduction
   D. Cast immobilization

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16. A patient present with an open fracture tibia, the wound measuring 1.5cm. Which Gustillo Anderson grade is this?
    A. Grade I
    B. Grade IIIA
    C. Grade II
    D. Grade IIIB
17. Spiral fracture is due to:
    A. Blunt trauma
    B. Direct impact
    C. Twisting forces
    D. Axial compression
18. Which of the following is a systemic cause of pathological fractures?
    A. Infection
    B. Osteoporosis
    C. Poor blood supply
    D. Soft tissue interposition
19. Which of the following is a principle of immobilization of long bone fractures?
    A. Immobilize fractured bone only
    B. Immobilize proximal joint
    C. Immobilize both proximal and distal joints
    D. Immobilize joint involved in the fracture
20. In which of the following injuries is skull traction indicated?
    A. Fracture spine of scapula
    B. Fracture of the clavicle
    C. Fracture of the cervical spine
    D. Fracture of the skull bone
21. A patient with spine injury and suspected to have spinal cord compression, which is the best investigation that should be requested?
    A. CT scan
    B. ESR
    C. MRI
    D. X-ray
22. Which of the internal fixation method best suits fracture of the neck of femur?
    A. Intramedullary nail
    B. Tension band wires
    C. Kitschier wires fixation
    D. Dynamic hip screw

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23. Which of the following dislocations of the hip is associated with injury of the sciatic nerve?
   A. Anterior dislocation
   B. Lateral dislocation
   C. Central dislocation
   D. Posterior dislocation
24. A 20-year-old patient fell and sustained fracture surgical neck of humerus. After 4 weeks the immobilization was removed and was unable to abduct the shoulder. The most likely caused is?
   A. Injury to suprascapular nerve
   B. Failure of fractured fragments to unite
   C. Axillary nerve injury
   D. Dorsal scapular nerve injury
25. External fixation is particularly used for
   A. Simple transverse fracture
   B. Closed fracture
   C. Open fractures
   D. Fractures without soft tissue damage
26. Compartment syndrome is treated by:
   A. IV fluids
   B. Fasciotomy
   C. Bicarbonates
   D. Analgesia
27. Dashboard injuries results in:
   A. Posterior hip dislocation
   B. Central hip dislocation
   C. Anterior hip dislocation
   D. Fracture neck femur
28. Gunstock deformity is due to?
   A. Fracture end of radius
   B. Lateral condyle fracture of humerus
   C. Supracondylar fracture of humerus
   D. Fracture of 2nd metacarpal bone
29. Galeazzi fracture is found in which of following?
   A. Lower end of radius
   B. Upper end of radius
   C. Upper end of ulna
   D. Lower end of ulna
30. Overwhelming infection and resulting vasodilation can lead to the following type of shock:
   A. Anaphylactic
   B. Cardiogenic
   C. Hypovolemic
   D. Septic

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31. Which one of the following muscles is part of the rotator cuff of the shoulder?
   A. Deltoid
   B. Teres major
   C. Pectoralis major
   D. Subscapularis
32. Which is the last stage in fracture healing?
   A. Hematoma
   B. Callus formation
   C. Consolidation
   D. Remodeling
33. Un-displaced fracture neck humerus in elderly is best treated by:
   A. Arthroplasty
   B. Above elbow backslab
   C. Sugar tong splint
   D. Analgesics and armsling
34. The most commonly fractured carpal bone is:
   A. Trapezium
   B. Trapezoid
   C. Scaphoid
   D. Lunate
35. The term bumper fracture refers to:
   A. A fracture of both the malleoli
   B. Fracture of the greater trochanter of femur
   C. Fracture of the shaft of humerus
   D. Fracture of the lateral tibial plateau
36. Wedge compression fractures are commonly seen in:
   A. Skull bones
   B. Tarsal bones
   C. Vertebral bodies
   D. Pelvic bones
37. A Potts fracture is a type of fracture found in the
   A. Foot
   B. Wrist
   C. Ankle
   D. Patella
38. A 5-year old has a fracture of the right upper arm. The x-ray should that one side of the bone is bend while the other is broken. This is known as
   A. Spiral
   B. Oblique
   C. Green stick
   D. Transverse

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39. Gutter splint is a treatment method for fractures of:
    A. Humeral bones
    B. Metacarpals bones
    C. Tibia fibula fractures
    D. Tarsal bones
40. Displaced fracture of the patella bone is treated by:
    A. Full length pop cast
    B. Plaster cylinder
    C. Tension wire bands
    D. Intra-medullary nail

SECTION 2: SHORT ANSWER QUESTIONS (40 MARKS)

41. A patient sustains a deep cut on his leg after an assault with sharp object. Highlight the recommended treatment for this patient. **(5 marks)**
42. Highlight clinical presentation of a patient with Colle's fracture **(5 marks)**
43. State five (5) systemic factors that affect bone healing in a patient with a closed fracture **(5 marks)**
44. Outline strategies that should be put in place to prevent road traffic accident in Kenya **(5 marks)**
45. Describe the clinical diagnosis of a patient with fracture mid tibia and suspected compartment syndrome. **(5 marks)**
46. Describe five (5) common soft tissue injuries that can occur at the knee **(5 marks)**
47. Elaborate five (5) causes of spine injuries **(5 marks)**
48. A patient comes to the clinic and following clinical assessment you suspect anterior shoulder dislocation. Enumerate clinical presentation of this patient. **(5 marks)**

SECTION 3: LONG ANSWER QUESTIONS (20 MARKS)

49. Discuss the classification and management of a patient diagnosed with epiphyseal injury of the tibia **(20 marks)**

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